Quiz 2: Detailed Reference Solutions

Economics and Computation, 2026 Fall

Based on the revised `quiz_02.pdf`

Problem 1. Dubins–Spanier on a small instance

The density rows are $f_1 = (3, 0, 0)$, $f_2 = (0, 2, 1)$, and $f_3 = (2, 0, 1)$. Each agent values the whole cake at 1. At every competitive stage, the marking target is the fixed value $1/3$.

(a) Marks and final pieces

Stage 1: remaining cake $[0, 1]$. All three agents are active. Their leftmost marks are

$$\begin{aligned} V_1([0, y_1]) = 3y_1 = \tfrac{1}{3} \quad &\Longrightarrow \quad y_1 = \tfrac{1}{9}, \\ V_2([0, y_2]) = 2(y_2 - \tfrac{1}{3}) = \tfrac{1}{3} \quad &\Longrightarrow \quad y_2 = \tfrac{1}{2}, \\ V_3([0, y_3]) = 2y_3 = \tfrac{1}{3} \quad &\Longrightarrow \quad y_3 = \tfrac{1}{6}. \end{aligned}$$

Since $1/9 < 1/6 < 1/2$, agent 1 has the leftmost mark. She receives $A_1 = [0, 1/9]$ and leaves.

Stage 2: remaining cake $[1/9, 1]$. Agents 2 and 3 remain. Agent 2 values $[1/9, 1/3]$ at zero, so her mark is still $1/2$. Agent 3's mark is reached within the first third:

$$\begin{aligned} 2(y_2 - \tfrac{1}{3}) = \tfrac{1}{3} \quad &\Longrightarrow \quad y_2 = \tfrac{1}{2}, \\ 2(y_3 - \tfrac{1}{9}) = \tfrac{1}{3} \quad &\Longrightarrow \quad y_3 = \tfrac{1}{9} + \tfrac{1}{6} = \tfrac{5}{18}. \end{aligned}$$

Since $5/18 < 1/2$, agent 3 receives $A_3 = [1/9, 5/18]$ and leaves.

Final step. Agent 2 receives all remaining cake, $A_2 = [5/18, 1]$. The procedure requires no mark from the final agent.

The marks and recipients can be summarized as follows:

Stage	Remaining cake	y_1	y_2	y_3	Recipient
1	$[0, 1]$	$\frac{1}{9}$	$\frac{1}{2}$	$\frac{1}{6}$	1
2	$[\frac{1}{9}, 1]$	inactive	$\frac{1}{2}$	$\frac{5}{18}$	3
Final	$[\frac{5}{18}, 1]$	inactive	not needed	inactive	2

Thus

$$\boxed{A_1 = [0, \tfrac{1}{9}], \qquad A_2 = [\tfrac{5}{18}, 1], \qquad A_3 = [\tfrac{1}{9}, \tfrac{5}{18}].}$$

Endpoints have value zero and may be assigned to either adjacent piece. No tie occurs at either competitive stage.

Problem 1, continued

(b) Value table, proportionality, and envy

The entry in row i, column j is $V_i(A_j)$, so each row evaluates *all three pieces using the same agent's density.*

	A_1	A_2	A_3
V_1	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$
V_2	0	1	0
V_3	$\frac{2}{9}$	$\frac{4}{9}$	$\frac{1}{3}$

For completeness, the entries are obtained as follows. Agent 1 values only the first third:

$$V_1(A_1) = 3\left(\frac{1}{9} - 0\right) = \frac{1}{3},$$
$$V_1(A_2) = 3\left(\frac{1}{3} - \frac{5}{18}\right) = \frac{1}{6},$$
$$V_1(A_3) = 3\left(\frac{5}{18} - \frac{1}{9}\right) = \frac{1}{2}.$$

Agent 2 values the first third at zero. Both A_1 and A_3 lie within that interval, whereas A_2 contains all of the cake she values:

$$V_2(A_1) = 0,$$
$$V_2(A_2) = 2\left(\frac{2}{3} - \frac{1}{3}\right) + 1\left(1 - \frac{2}{3}\right) = 1,$$
$$V_2(A_3) = 0.$$

For agent 3, the density is 2 on the first third, 0 on the second third, and 1 on the final third:

$$V_3(A_1) = 2\left(\frac{1}{9} - 0\right) = \frac{2}{9},$$
$$V_3(A_2) = 2\left(\frac{1}{3} - \frac{5}{18}\right) + 1\left(1 - \frac{2}{3}\right) = \frac{1}{9} + \frac{1}{3} = \frac{4}{9},$$
$$V_3(A_3) = 2\left(\frac{5}{18} - \frac{1}{9}\right) = \frac{1}{3}.$$

Proportionality. All total values equal 1, and the agents' own values are

$$V_1(A_1) = \frac{1}{3}, \qquad V_2(A_2) = 1, \qquad V_3(A_3) = \frac{1}{3}.$$

Each is at least $1/3$, so the allocation is **proportional**.

Envy. Agent 1 envies agent 3 because $1/3 < 1/2$, but does not envy agent 2. Agent 2 values her own piece at 1 and both other pieces at zero, so she envies nobody. Agent 3 envies agent 2 because $1/3 < 4/9$, but does not envy agent 1, whose piece she values at $2/9$. Therefore the allocation is **not envy-free**, and the complete set of ordered envy pairs is

$$\boxed{\{(i,j) : V_i(A_i) < V_i(A_j)\} = \{(1,3),(3,2)\}.}$$

Problem 2. Optimal proportional cake cutting

(a) Utilities and the linear program

The intervals L and R each have length $1/2$. Since x and y are *fractions of these lengths*, agent 1 receives length $x/2$ from L and length $y/2$ from R.

Agent 1 has densities 2 on L and 0 on R, so

$$\boxed{V_1(A_1) = 2\frac{x}{2} + 0\frac{y}{2} = x.}$$

Agent 2 receives the remaining lengths $(1-x)/2$ and $(1-y)/2$. Her densities are $3/2$ on L and $1/2$ on R, giving

$$\boxed{V_2(A_2) = \frac{3}{2}\frac{1-x}{2} + \frac{1}{2}\frac{1-y}{2} = \frac{3}{4}(1-x) + \frac{1}{4}(1-y) = 1 - \frac{3}{4}x - \frac{1}{4}y.}$$

Both agents value the whole cake at 1. With two agents, proportionality therefore requires each agent to receive value at least $1/2$:

$$x \geq \frac{1}{2}, \qquad 1 - \frac{3}{4}x - \frac{1}{4}y \geq \frac{1}{2}.$$

The utilitarian social welfare is

$$\mathrm{SW}(A) = x + 1 - \frac{3}{4}x - \frac{1}{4}y = 1 + \frac{1}{4}x - \frac{1}{4}y.$$

Consequently, a correct linear program is

$$\boxed{\begin{array}{ll} \text{maximize} & 1 + \frac{1}{4}x - \frac{1}{4}y \\ \text{subject to} & x \geq \frac{1}{2}, \\ & 3x + y \leq 2, \\ & 0 \leq x \leq 1, \quad 0 \leq y \leq 1. \end{array}}$$

Equivalently, one may maximize $(x-y)/4$ or $x-y$, because deleting a constant and multiplying the objective by a positive constant preserve its maximizers.

Additional verification: solving the LP

The question asks only for the formulation. Its optimal solution can also be obtained directly. Every feasible point satisfies

$$x \leq \frac{2-y}{3},$$

and therefore

$$\mathrm{SW}(A) \leq 1 + \frac{2-y}{12} - \frac{y}{4} = \frac{7}{6} - \frac{y}{3} \leq \frac{7}{6}.$$

The point $(x, y) = (2/3, 0)$ is feasible and attains this bound. Hence

$$\boxed{x^* = \frac{2}{3}, \qquad y^* = 0, \qquad \mathrm{SW}^* = \frac{7}{6}.}$$

One realizing allocation is $A_1 = [0, 1/3]$ and $A_2 = [1/3, 1]$. Its utilities are $2/3$ and $1/2$, respectively, so both proportionality constraints hold.

Problem 3. Checking EF, EF1, and EFX

The allocation is $A_1 = \{g_1\}$ and $A_2 = \{g_2, g_3, g_4\}$. The values relevant to comparing the two bundles are

	A_1	A_2
v_1	4	$5+2+1=8$
v_2	2	$4+3+2=9$

(a) Is the allocation EFX?

$$\boxed{\text{No.}}$$

This is the complete response requested for part (a). The calculations below also show why EFX fails.

(b) Largest α for which the allocation is α-EFX

For agent 1's comparison with agent 2, the three possible removals give

Removed good g	$v_1(A_2 \setminus \{g\})$	Required inequality
g_2	$2+1=3$	$4 \geq 3\alpha$
g_3	$5+1=6$	$4 \geq 6\alpha$
g_4	$5+2=7$	$4 \geq 7\alpha$

All three inequalities must hold. Thus

$$\alpha \leq \min\left\{\frac{4}{3}, \frac{4}{6}, \frac{4}{7}\right\} = \frac{4}{7}.$$

In particular, ordinary EFX ($\alpha = 1$) fails, since removing g_4 leaves a bundle worth 7 to agent 1, exceeding her own value 4.

For agent 2's comparison with agent 1, the only removal is g_1, leaving the empty bundle. The requirement is

$$v_2(A_2) = 9 \geq \alpha v_2(\varnothing) = 0,$$

which imposes no further restriction. Comparisons with $i = j$ also impose no restriction: nonnegativity and $0 \leq \alpha \leq 1$ imply $v_i(A_i) \geq \alpha v_i(A_i \setminus \{g\})$.

At $\alpha = 4/7$, the three constraints for agent 1 become

$$4 \geq \frac{12}{7}, \qquad 4 \geq \frac{24}{7}, \qquad 4 = \frac{4}{7} \cdot 7.$$

Hence $4/7$ is feasible, while every larger α fails on removal of g_4. Therefore

$$\boxed{\alpha_{\max} = \frac{4}{7}.}$$

Supplement: EF and EF1

The allocation is not EF because $v_1(A_1) = 4 < 8 = v_1(A_2)$. It is EF1: removing g_2 from A_2 leaves value $3 \leq 4$ for agent 1; agent 2 does not envy agent 1 in the first place. EF1 requires a suitable removal to exist, whereas EFX requires the inequality to hold for every removal.

Problem 4. Computing and attaining maximin shares

(a) Computing $\mu_1^2(M)$ and $\mu_2^2(M)$

For agent i, the maximin share is the largest minimum bundle value she can guarantee by partitioning all goods into two bundles:

$$\mu_i^2(M) = \max_{(P_1,P_2) \text{ partition of } M} \min\{v_i(P_1), v_i(P_2)\}.$$

Agent 1: lower bound. Choose $P_1 = \{g_1\}$ and $P_2 = \{g_2, g_3\}$. Their values are 8 and $7 + 6 = 13$. Consequently,

$$\mu_1^2(M) \geq \min\{8, 13\} = 8.$$

Agent 2: lower bound. Choose $P_1 = \{g_2\}$ and $P_2 = \{g_1, g_3\}$. Their values are 8 and $7 + 5 = 12$. Consequently,

$$\mu_2^2(M) \geq \min\{8, 12\} = 8.$$

Matching upper bounds for both agents. Any partition of three goods into two bundles has a bundle containing at most one good. If that bundle is empty, its value is 0. If it is a singleton, its value is at most 8 for either agent, because neither agent values any individual good above 8. Thus the smaller bundle value is always at most 8:

$$\min\{v_i(P_1), v_i(P_2)\} \leq 8 \qquad (i = 1, 2).$$

Maximizing over all partitions preserves this bound, proving

$$\boxed{\mu_1^2(M) = \mu_2^2(M) = 8.}$$

As a direct check, the three unordered nonempty partitions give

Partition	Agent 1's minimum	Agent 2's minimum
$\{g_1\} \mid \{g_2, g_3\}$	$\min(8, 13) = 8$	$\min(7, 13) = 7$
$\{g_2\} \mid \{g_1, g_3\}$	$\min(7, 14) = 7$	$\min(8, 12) = 8$
$\{g_3\} \mid \{g_1, g_2\}$	$\min(6, 15) = 6$	$\min(5, 15) = 5$

(b) A complete MMS allocation

Take

$$\boxed{A_1 = \{g_1\}, \qquad A_2 = \{g_2, g_3\}.}$$

These bundles are disjoint and exhaust the goods. The two guarantees are

$$v_1(A_1) = 8 \geq \mu_1^2(M) = 8, \qquad v_2(A_2) = 8 + 5 = 13 \geq \mu_2^2(M) = 8.$$

Another valid allocation is $A_1 = \{g_1, g_3\}$ and $A_2 = \{g_2\}$, with utilities 14 and 8. Each agent's MMS is computed using her own valuation; the partitions used to establish the two MMS values need not coincide.

Problem 5. The p-mean and Nash social welfare

For fixed $u_i > 0$, define

$$W_p = \left(\frac{1}{n} \sum_{i=1}^{n} u_i^p \right)^{1/p}, \qquad p \in \mathbb{R} \setminus \{0\}.$$

We prove that the two-sided limit at zero is the geometric mean.

Step 1: take logarithms. Since all utilities are positive, u_i^p and W_p are positive for every real $p \neq 0$. Therefore

$$\log W_p = \frac{\log \left(\frac{1}{n} \sum_{i=1}^{n} u_i^p \right)}{p}.$$

Substituting $p = 0$ into this expression would give $0/0$; it does not directly evaluate the limit.

Step 2: define a differentiable function at zero. Let

$$S(p) = \frac{1}{n} \sum_{i=1}^{n} e^{p \log u_i} = \frac{1}{n} \sum_{i=1}^{n} u_i^p, \qquad h(p) = \log S(p).$$

The function S is positive and differentiable on all of $\mathbb{R}$. Thus h is differentiable as well, and

$$S(0) = 1, \qquad h(0) = 0.$$

For $p \neq 0$ we have

$$\log W_p = \frac{h(p) - h(0)}{p - 0}.$$

The definition of the derivative now gives

$$\lim_{p \to 0} \log W_p = h'(0).$$

Step 3: compute the derivative. Differentiating the finite sum,

$$S'(p) = \frac{1}{n} \sum_{i=1}^{n} u_i^p \log u_i,$$

and hence

$$h'(0) = \frac{S'(0)}{S(0)} = \frac{1}{n} \sum_{i=1}^{n} \log u_i.$$

Equivalently, this computation may be presented as an application of l'Hôpital's rule to $\log S(p)/p$.

Step 4: exponentiate. By continuity of the exponential function,

$$\boxed{\lim_{p \to 0} W_p = \exp \left(\frac{1}{n} \sum_{i=1}^{n} \log u_i \right) = \left(\prod_{i=1}^{n} u_i \right)^{1/n}.}$$

The derivative argument applies from both sides of zero, including negative p. The factor $1/n$ in the definition of W_p yields the geometric mean, which is the normalized Nash social welfare.